

# Math AA HL at KCA - Chapter 17 Notes

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## Contents

|          |                                   |          |
|----------|-----------------------------------|----------|
| <b>1</b> | <b>Basic Rules of Integration</b> | <b>3</b> |
| <b>2</b> | <b>Reverse Chain Rule</b>         | <b>4</b> |
| <b>3</b> | <b>Trigonometric Integrals</b>    | <b>5</b> |
| <b>4</b> | <b>Integration by Parts</b>       | <b>6</b> |
| <b>5</b> | <b>Volume of Revolution</b>       | <b>7</b> |

## 1 Basic Rules of Integration

$$\int u' \cdot u^n \, dx = \frac{u^{n+1}}{n+1} + C$$

$$\int \frac{u'}{u} \, dx = \ln |u| + C$$

$$\int u' \cdot e^u \, dx = e^u + C$$

$$\int u' \cdot a^u \, dx = \frac{a^u}{\ln a} + C$$

## 2 Reverse Chain Rule

When you notice that a function is multiplied with its derivative, it's likely the reverse chain rule.

### 3 Trigonometric Integrals

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \tan x \, dx = -\ln |\cos x| + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int -\frac{1}{\sqrt{a^2 - x^2}} \, dx = \arccos\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C \text{ with } x < a$$

## 4 Integration by Parts

$$\int uv' \, dx = uv - \int vu' \, dx$$

## 5 Volume of Revolution

If we revolve the curve  $y = f(x)$  about the  $x$ -axis, the volume of the solid generated is given by

$$V = \int_a^b \pi y^2 \, dx$$